

Total No. of Printed Pages: 2

SUBJECT CODE NO: RNP-8002
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Civil Engg.) (NEP) Sem - III
Examination May/June-2026
Fluid Mechanics

[Time: 3:00 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION-A**Q.1 Solve any 10 following questions****20**

1. The unit of dynamic viscosity is:
a) m^2/s b) Ns/m^2 c) kg/m^3 d) N/m^3
2. A fluid which has no resistance to shear stress is called:
a) Real fluid b) Ideal fluid
c) Newtonian fluid d) Non-Newtonian fluid
3. The value of Reynolds number for laminar flow is:
a) $Re > 4000$ b) $Re < 2000$
c) $Re = 3000$ d) $Re > 10000$
4. The device used to measure flow velocity is:
a) Venturimeter b) Pitot tube
c) Orifice meter d) Manometer
5. The equation representing conservation of mass is:
a) Bernoulli's equation b) Euler's equation
c) Continuity equation d) Momentum equation
6. Pressure at a point in a static fluid is:
a) Different in all directions b) Same in all directions
c) Zero d) Depends on area
7. State Pascal's Law.
8. Define capillarity.
9. What is metacentric height?
10. Define surface tension.
11. Define cavitation in pumps.
12. Define hydraulic gradient line (HGL).
13. What is specific energy in open channel flow?

SECTION-B

- Q.2** a) Explain rheological diagram and classify fluids based on it. **05**
 b) Explain hydraulic gradient line (HGL) and total energy line (TEL). **05**
- Q.3** a) State and explain Pascal's Law with applications. **05**
 b) Determine the total pressure on a circular plate of diameter 1.5 m which is placed vertically in water in such a way that the centre of the plate is 3.5 m below the free surface of water. Find the position of the centre of pressure. **05**
- Q.4** a) Explain major and minor losses in pipes. **05**
 b) State and derive Bernoulli's equation. **05**
- Q.5** a) Explain surface tension and capillarity with examples **05**
 b) Explain hydraulic jump and its applications. **05**
- Q.6** a) Explain model and prototype relations. **05**
 b) Water flows through a pipe of diameter 0.3 m and length 500 m with velocity 2 m/s . Find the loss of head due to friction. Take friction factor $f = 0.02$. **05**
- Q.7** a) Explain centrifugal pump working with diagram. **05**
 b) Explain siphon with neat sketch. **05**